

Vitrectomy and Vitrector Port Needle Biopsy of Choroidal Melanoma for Gene Expression Profile Testing Immediately before Brachytherapy

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Purpose: Transvitreal and transscleral needle biopsy can result in complications including vitreous hemorrhage and retinal detachment. This study evaluated a technique using 25-gauge vitrectomy as an adjunct to needle biopsy immediately before brachytherapy to minimize these complications and preserve good visual acuity.

Design: Retrospective, observational case series.

Participants: Fifty-seven consecutive eyes of 57 patients with treatment-naïve medium choroidal melanomas without extraocular extension from July 2012 through September 2015.

Methods: Fifty-seven consecutive eyes of 57 patients with a clinically diagnosed choroidal melanoma underwent complete 25-gauge posterior vitrectomy followed by transvitrector port fine-needle aspiration biopsy of the tumor immediately before implantation of a radioactive iodine 125 plaque as treatment for the tumor. Cytopathologic analysis was not performed on the tumor aspirates in this study.

Main Outcome Measures: Best-corrected postoperative visual acuity, postoperative complications of the reported technique, implantation tumor development, local tumor recurrence, presence of metastatic disease after surgery, and sufficiency of the tumor aspirates obtained by the reported technique for successful gene expression profile testing and prognostic classification.

Results: Mean preoperative and postoperative visual acuities were similar (20/60 vs. 20/80, respectively). Mean tumor thickness was 5.0 mm (range, 2.5–10 mm) and mean tumor basal diameter was 13.1 mm (range, 7–22 mm). Only 1 of 57 eyes (1.8%) showed a transient vitreous hemorrhage, biopsy yield was 100% for genetic analysis, and no patients showed recurrence or implantation tumor at the vitrector site.

Conclusions: Combined 25-gauge posterior vitrectomy and 25-gauge trans-vitrector port needle aspiration biopsy immediately before brachytherapy is excellent for obtaining tumor aspirate for gene expression profiling while controlling for hemostasis, resulting in few complications. *Ophthalmology* 2017;124:1377-1382 © 2017 by the American Academy of Ophthalmology



Supplemental video available at www.aaojournal.org.

Of the primary intraocular malignancies, uveal melanoma is the most common with an annual incidence of approximately 6 cases per 1 million.^{1–3} Even with advancements in the treatment of choroidal melanomas, approximately 50% of patients die within 25 years secondary to metastases.^{4,5} Although the diagnosis of classic choroidal melanoma usually can be made based on clinical examination and noninvasive test results, fine-needle aspiration biopsy (FNAB) of the tumor as an additional surgical intervention in combination with sight-preserving brachytherapy, in lieu of en bloc tumor resection or enucleation, has emerged as one method accepted for determining metastatic potential. Other methods for determining metastatic potential include 2010 tumor, node, metastasis classification based on tumor characteristics as well as cytopathologic classification of

tumor aspirates. With the use of genetic expression profiling (GEP), uveal melanomas can be classified as having low metastatic risk (class 1) or as having a substantially higher risk of clinical metastases developing and leading to death as a result of metastatic disease (class 2) using a polymerase chain reaction–based 15-gene assay.⁶ This RNA-based GEP test (Castle Biociences, Inc, Friendswood, TX) has been shown in several studies to be more accurate than the clinical, pathologic, and chromosomal prognostic factors in predicting the likelihood of metastases.^{7–11} In addition, it requires a smaller amount of biopsy material for analysis.⁶

Currently, FNAB via a transscleral or transvitreal (without concomitant vitrectomy) route is used to biopsy choroidal melanomas.¹² The transvitreal approach necessitates perforation of the retina and puncturing of the